

Quiz 4

● Graded

Student

Samyak Sanghvi

Total Points

4 / 4 pts

Question 1

Q1

2 / 2 pts

✓ + 1 pt w is even $\Rightarrow W$ is real

✓ + 1 pt multiplication with real number doesn't change phase

+ 0 pts No/Wrong answer

Question 2

Q2

2 / 2 pts

✓ + 2 pts Smoothed falloff filter

+ 0 pts No/Wrong answer

Name: Samyak Sanghvi

COL7(6)83 Quiz 4

Entry no.: 2023CS10807

29 August, 2025

1. An image f is convolved with a kernel w having even symmetry, i.e. $w(x, y) = w(-x, -y)$. What happens to the phase of its Fourier transform, F ?

$$\text{let } g = f * w$$

Since w is even ($W \Leftrightarrow w$) W is real and even.

$$\therefore G = F \cdot W$$

Since W is real, just acts as a scaling factor

$\Rightarrow$ no change in phase but norm will be

$$|F| \cdot |W| \quad \text{---} \quad |F|$$

2. I have an image affected by a periodic interference pattern, which I attempt to remove by applying an ideal notch filter, i.e. $H(u, v)$ is exactly 0 near the undesirable frequencies and exactly 1 elsewhere. I find that this removes the interference pattern but creates some extraneous intensity oscillations. What should I change to get a better result?

Instead of using an ideal filter, which causes ringing, you can simply use a smoother filter like gaussian filter or butterworth to get better results. Shift the mean of these filters to be close to mean of $H(u, v)$ to work better.

$$\text{eg. } H'(u, v) = \frac{1}{1 + \left(\frac{D(u-u_0, v-v_0)}{D_0} \right)^{2n}}$$